# IBM - BUILDING RAG APPLICATIONS

## SUMMARY

- Retrieval-Augmented Generation (RAG) is a machine-learning technique that integrates information retrieval with generative AI to produce accurate and context-aware responses.

- RAG enhances Large Language Models (LLMs) by integrating external or domain-specific knowledge without retraining. This helps LLMs generate more accurate and contextually relevant responses for specialized queries, such as a company's mobile policy.

- RAG consists of two main components: the Retriever, which extracts relevant data from a knowledge base, and the Generator, which uses the retrieved information to generate responses in natural language.

- The RAG process comprises four steps: Text Embedding, Retrieval, Augmented Query Creation, and Model Generation.

- Text Embedding converts user prompts and knowledge base documents into high-dimensional vectors using AI models such as BERT or GPT.

- Retrieval matches the user query with similar vectors from the knowledge base to retrieve relevant information.

- Augmented Query Creation combines retrieved content with the user prompt.

- [ ] Model Generation uses the created augmented query to generate a response using the content from the knowledge base.

- [ ] Prompt encoding converts a text-based prompt into a numerical representation that an LLM can process. It uses Token Embedding and Vector Averaging to break documents into smaller text chunks, convert them into vectors, and index them in a vector database.

- [ ] Distance Metrics measure similarity between user queries and document vectors using methods such as dot product (magnitude-based) or cosine similarity (direction-based).

- [ ] RAG is an efficient response generation technique. It retrieves the most relevant text chunks from the knowledge base to augment the model's knowledge and produce an informed response. This ensures responses are accurate, domain-specific, and up-to-date.

- [ ] RAG allows chatbots to provide specialized answers by integrating relevant external knowledge, making them more reliable for industry-specific or confidential topics.

# CHEAT - SHEET

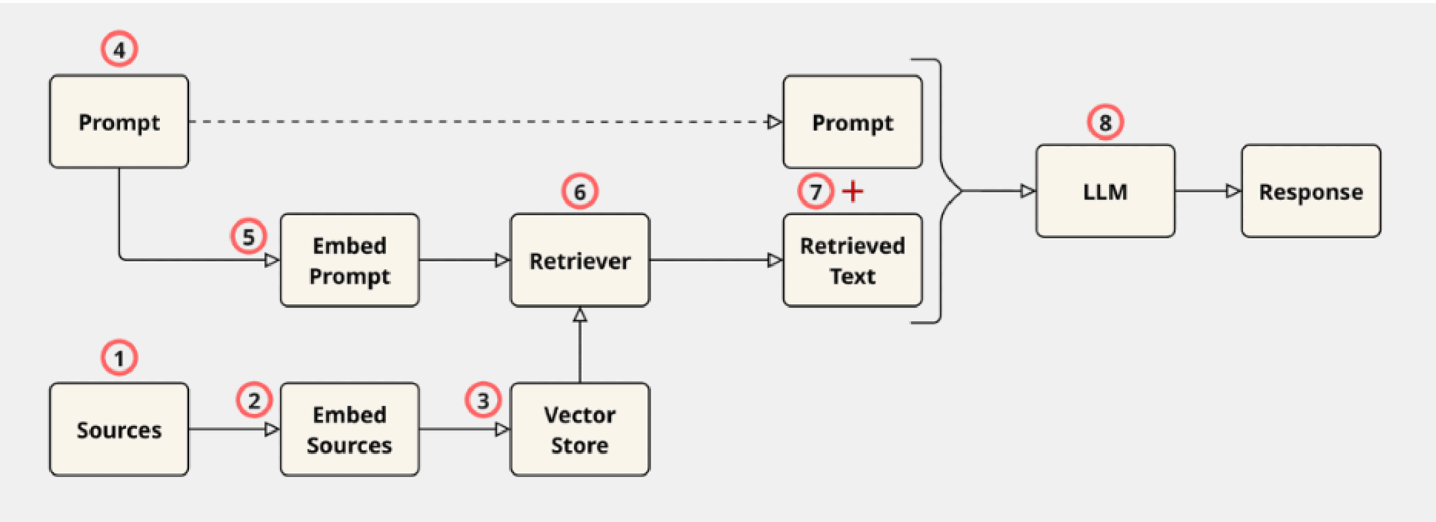
- [ ] RAG (Retrieval-Augmented Generation) is an AI technique that combines relevant information from external data sources with generative models to produce accurate and contextually rich responses.

- [ ] Non-RAG systems have a simple Prompt → LLM → Response pipeline. These systems may generate inaccurate, outdated, or fabricated responses.

- [ ] RAG aims to augment the prompt with relevant text from an external data store. This enhances response quality by providing the LLM with relevant information, provides up-to-date information to the LLM, and allows the user to verify the sources on which the response is generated.

- [ ] Using models with long context lengths instead of RAG requires users to provide the necessary contextual information within the prompt. This approach has limited capacity, may lead to response quality issues when relevant facts are buried within large amounts of irrelevant text, and increases processing time and costs. RAG mitigates these challenges, either partially or fully.

- [ ] The following diagram illustrates the RAG process for a basic RAG system:




- [ ] Large sources are usually split into smaller chunks prior to or at the embedding step

- [ ] Embedding involves tokenization (splitting text into tokens and assigning unique numerical IDs to each token) followed by passing the token IDs through a neural network to produce fixed-length numeric vectors, one for each text chunk.

- [ ] Vectors are stored in vector databases that specialize in efficiently processing vector data. Examples of vector databases include ChromaDB, Faiss, and Milvus.

- [ ] User's prompts are embedded using the same model as the documents.

- [ ] Retrievers retrieve the most similar chunks to the embedded prompt from the vector store. The most similar chunks are found through a similarity search. Methods of determining similarity include cosine similarity, which only compares the angles of vectors, and the dot product, which compares their angles and magnitudes.